

Bill of Materials — V10a_final_h45 (g/cell and kg/kWh)

Component	Material	Mass g/cell	kg/kWh	Source
Positive electrode active material	NMC811	14.5881	0.7928	96 wt% of cathode solids — literature default (set has no AM-fraction key)
Positive electrode conductive additive	carbon black	0.3039	0.0165	2 wt% — literature default (no parameter)
Positive electrode binder	PVDF	0.3039	0.0165	2 wt% — literature default (no parameter)
Negative electrode active material	graphite	9.0473	0.4917	96 wt% — literature default (no parameter)
Negative electrode conductive additive	carbon black	0.0942	0.0051	1 wt% — literature default (no parameter)
Negative electrode binder	CMC/SBR	0.2827	0.0154	3 wt% — literature default (no parameter)
Separator	polyolefin	0.2161	0.0117	r6_final_energy_dfn.json:layer_kg_m2.separator x area
Electrolyte	EC/EMC + LiPF6	7.9810	0.4338	pore volume x 1.2 g/cm3 (literature density, annotated)
Positive current collector	Al foil	2.2183	0.1206	r6_final_energy_dfn.json:layer_kg_m2.positive_cc x area
Negative current collector	Cu foil	5.5212	0.3001	r6_final_energy_dfn.json:layer_kg_m2.negative_cc x area
Enclosure	—	Not modeled	Not modeled	Not modeled
Tabs	—	Not modeled	Not modeled	Not modeled
Total (layer solids, contract caliber)	—	32.5757	1.7704	r6_final_energy_dfn.json:mass_kg
Total incl. electrolyte estimate	—	40.5568	2.2042	electrolyte density = literature value 1.2 g/cm3
Rated energy	—	18.4000	Wh	r6_final_energy_dfn.json:energy_wh